Seonghwan Seo

Ph.D. Candidate | Korea Advanced Institute of Science and Technology
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RESEARCH INTERESTS

Generative Modeling
Generative Flow Networks (GFlowNets); diffusion and flow-based models

AI for Drug Discovery
Synthesis-aware molecular design; structure-based drug discovery

Biomolecular Design
Protein design; protein folding and biomolecular co-folding

EXPERIENCE

Team KAIST — South Korea
Biomolecular structure prediction — *Nov 2025 - Present*

- Developed AtlasFold for fast protein folding and co-folding
- Developed K-Fold for biomolecular complex structure prediction

Intelligent Chemistry Lab, KAIST Chemistry — Daejeon, South Korea
AI for Drug Discovery — *June 2019 - Present*
Supervisor: Prof. Woo Youn Kim

- Development of an unbiased drug-likeness scoring model
- Development of a deep learning framework for fully automated receptor-based pharmacophore modeling
- Research on automated drug discovery with synthesis-aware molecular generative modeling

HITS Inc. — Seoul, South Korea
AI Research Intern — *Dec. 2020 - Aug. 2022*
Manager: Dr. Jaechang Lim

- Development of a reaction template-based generative model for synthesizable molecular design with desired properties
- Development of a transformer-based model for synthesis pathway generation

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea
Ph.D. student in Chemistry — *Aug. 2022 - Feb. 2028 (expected)*
Supervisor: Woo Youn Kim

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea
B.S. in Chemistry & Computer Science (Double Major) — *Feb. 2018 - Aug. 2022*
Supervisor: Young Min Rhee
GPA: 3.99/4.3

PUBLICATIONS

First author

* indicates equal contribution; † indicates corresponding author

- Predicting conformational changes induced by biomolecular interactions [link]
  Yeongnam Bae, Jinheon Baek, Eugene Choi, Jeong Hun Choi, Hyeongyu Han, Hyosoon Jang, Woosung Jeon, Jaehyeong Jo, Dongki Kim, Hyeongwoo Kim, Jun Hyeong Kim, Jun Hyeong Kim, Shinwoo Kim, Taewon Kim, Youngchan Kim, Joongwon Lee, Junyup Lee, Seokhyun Moon, Minha Park, Calvin Samuel, Hyunjin Seo, Jisu Seo, **Seonghwan Seo**, Mingyeong Shin, Yunsu Shin, Wonho Zhung, Sungsoo Ahn, Dongsu Han, Sung Ju Hwang, Ho Min Kim, Woo Youn Kim, Gyu Rie Lee, Byung-Ha Oh
  bioRxiv preprint, 2026
  Authors listed in alphabetical order; served as Engineering Lead.
- Synthesis-aware generative design in trillion-scale chemical spaces for automated drug discovery [link]
  Seonghwan Seo, Yulseung Sung, Sang-Yeon Hwang, Mincheol Kang, Joonseong Lee, Samuele Bordi, Luka Raguz, Jihye Choi, Daniil Melnichenko, Wan Namkung, Sehan Lee, Jaechang Lim, Benedikt M. Wanner, Jung Min Han, Woo Youn Kim
  bioRxiv preprint, 2026

- AtlasFold: Protein structure prediction with metagenomic-scale language models [link]
  Seonghwan Seo†, Hyeongwoo Kim, Seokhyun Moon, Woo Youn Kim† Team KAIST
  bioRxiv preprint, 2026
- Boltz2ESI: Accurate Enzyme Specificity Prediction with Co-folding Foundation Model [link]
  Xiwei Cheng*, **Seonghwan Seo***, Jihang Chen, Songlin Jiang, Pengkang Guo, Wengong Jin†
  ICLR 2026 Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)
- Compositional Flows for 3D Molecule and Synthesis Pathway Co-design [link]
  Tony Shen*, **Seonghwan Seo***, Ross Irwin, Kieran Didi, Simon Olsson, Woo Youn Kim, Martin Ester†
  International Conference on Machine Learning (ICML), 2025
- Generative Flows on Synthetic Pathway for Drug Design [link]
  Seonghwan Seo, Minsu Kim, Tony Shen, Martin Ester, Jinkyoo Park, Sungsoo Ahn, Woo Youn Kim†
  International Conference on Learning Representations (ICLR), 2025
- PharmacoNet: deep learning-guided pharmacophore modeling for ultra-large-scale virtual screening [link]
  Seonghwan Seo, Woo Youn Kim†
  Chemical Science, 2024
- Molecular Generative Model via Retrosynthetically Prepared Chemical Building Block Assembly [link] [cover]
  Seonghwan Seo, Jaechang Lim†, Woo Youn Kim†
  Advanced Science, 2023 (selected for a journal cover)
- Drug-likeness scoring based on unsupervised learning [link]
  Kyunghoon Lee*, Jinho Jang*, **Seonghwan Seo***, Jaechang Lim†, Woo Youn Kim†
  Chemical Science, 2022

Co-author

- Atom-level Protein Representation Learning Improves Protein Structure Prediction [link]
  Taewon Kim, Hyosoon Jang, Hyunjin Seo, **Seonghwan Seo**, Hyeongwoo Kim, Wonho Zhung, Mingyeong Shin, Wooyoun Kim, Sungsoo Ahn†
  Advances in Neural Information Processing Systems (NeurIPS), 2026
- Generalized Multi-State Protein Design with AlphaFold3 [link]
  Xiwei Cheng, Pengkang Guo, **Seonghwan Seo**, Casper A. Goverde, Wengong Jin†
  ICLR 2026 Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)
- TacoGFN: Target Conditioned GFlowNet for Structure-based Drug Discovery [link]
  Tony Shen, **Seonghwan Seo**, Grayson Lee, Mohit Pandey, Jason R Smith, Artem Cherkasov, Woo Youn Kim, Martin Ester†
  Transactions on Machine Learning Research (TMLR), 2024

ACADEMIC SERVICE

- Organizer of the Generative AI and Biology (GenBio) at ICML 2026 in Seoul
- Reviewer (Conferences): ICLR (2025-2026), NeurIPS (2026)
- Reviewer (Journals): Nature Communications (2024)

PRESENTATIONS

Invited

- "Molecular Generative Model via Retrosynthetically Prepared Chemical Building Block Assembly"
  Oral workshop and tutorial at *AI-BIO, Artificial Intelligence Institute, Seoul National University*, Seoul, South Korea (Nov. 2022)

Contributed

- "Compositional Flows for 3D Molecule and Synthesis Pathway Co-design"
  Oral spotlight and poster presentation at *ICLR 2025 Workshop on AI for Accelerated Materials Discovery (AI4Mat)*, Singapore. (Apr. 2025)
- "Compositional Flows for 3D Molecule and Synthesis Pathway Co-design"
  Oral spotlight and poster presentation at *ICLR 2025 Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)*, Singapore. (Apr. 2025)
- "Generative Flows on Synthetic Pathway for Drug Design"
  Poster presentation at *ICLR 2025*, Singapore. (Apr. 2025)
- "Generative Flows on Synthetic Pathway for Drug Design"
  Poster presentation at *NeurIPS 2024 Workshop on AI for New Drug Modalities (AIDrugX)*, Vancouver, Canada. (Dec. 2024)

- "PharmacoNet: Accelerating Large-Scale Virtual Screening by Deep Pharmacophore Modeling"
  Poster presentation at *NeurIPS 2023 Workshop on New Frontiers of AI for Drug Discovery and Development (AI4D3)*, New Orleans, LA, USA. (Dec. 2023)
- "Molecular Generative Model via Retrosynthetically Prepared Chemical Building Block Assembly"
  Poster presentation at *2023 Accelerate Conference*, Toronto, Canada (Aug. 2023)

AWARDS & SCHOLARSHIPS

- **TTA President Award**
  Korea AI-Champion Competition, 2025
- **KAIST Alumni Scholarship**
  KAIST Alumni Association, 2021 - 2022
- **Younghoon Lee Scholarship**
  Department of Chemistry, KAIST, 2021
- **Model Student Awards**
  KAIST, 2021
- **Dean's List**
  College of Natural Science, KAIST, 2021
- **National Science & Technology Scholarship**
  Government of South Korea, 2020 - 2021
- **Dean's List**
  School in Freshman, KAIST, 2019

SKILLS

Languages
Korean (native), English (upper-intermediate)

Deep Learning Tools
Python, PyTorch, PyTorch Geometric, PyTorch Lightning, Numba, Pandas

Chemistry Tools
RDKit, OpenBabel, Gemmi, Vina, Smina, rDock